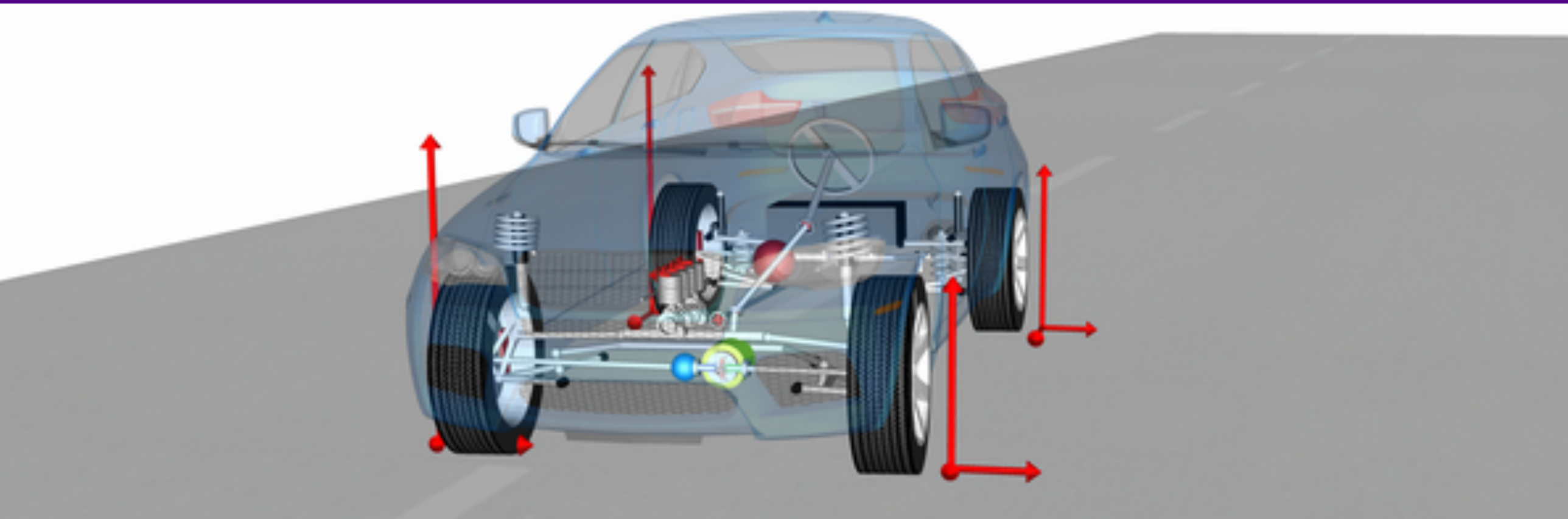
NYU

ROB-UY 3303

Robot Motion & Planning

Lecture 03A - Motion Modeling



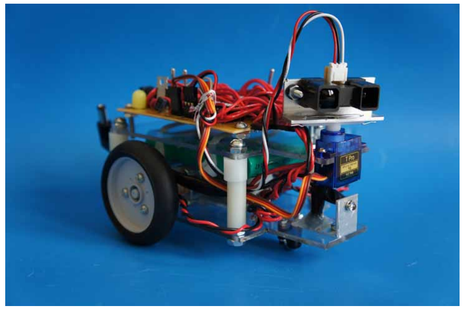
Agenda

1. Kinematics
2. Differential Drive Robot + Wheel Encoder
3. Differential Drive Robot + Wheel Velocity

Agenda

1. **Kinematics**
2. Differential Drive Robot + Wheel Encoder
3. Differential Drive Robot + Wheel Velocity

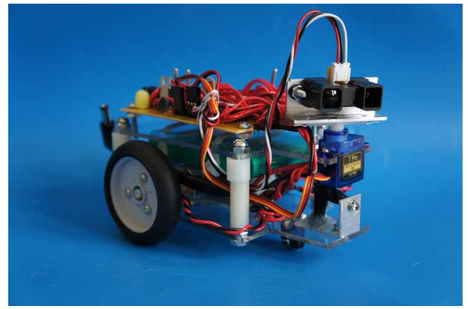
- **Motion Modeling**
  - An equation that relates the motion of an object to previous states and inputs
  - Usually starts with **kinematic** or **dynamic** differential equations



$$x_t = f(x_{t-1}, u_t)$$

- **Motion Modeling**

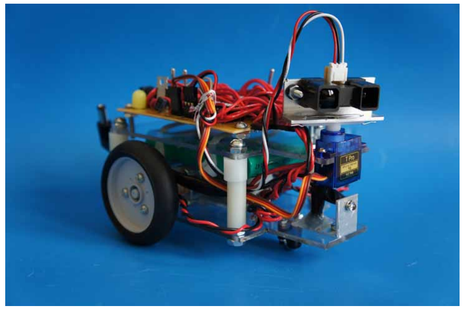
  - Can be discrete or continuous time.
  - Can be linear or non-linear



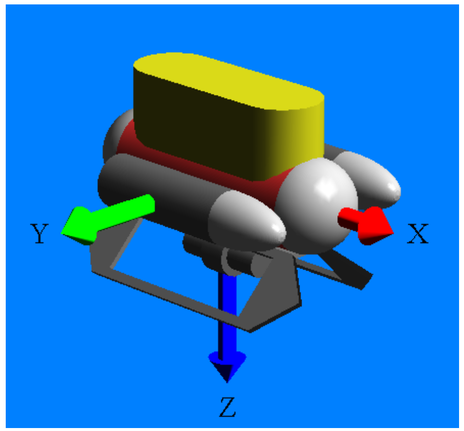
$$x_t = f(x_{t-1}, u_t)$$

$$\dot{x} = f(x, u)$$

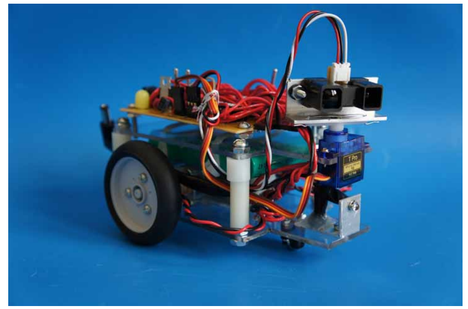
- **Kinematics**
  - The motion of an object as a function of geometry (position).



- **Dynamics**
  - The motion of an object as a function of forces.



- **Probabilistic K & D**
  - Our states can be represented as vectors of random variables.
  - Our motion models can be represented as conditional probabilities.



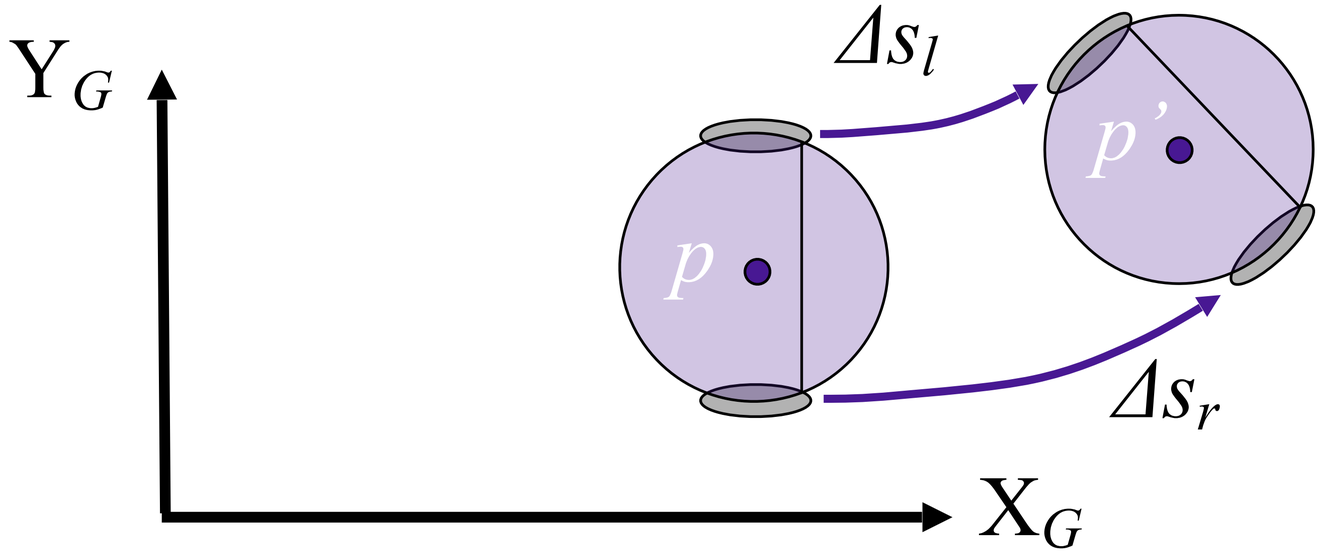
$$p(x_t \mid x_{t-1}, u_t)$$

$$p(\dot{x} \mid x, u)$$

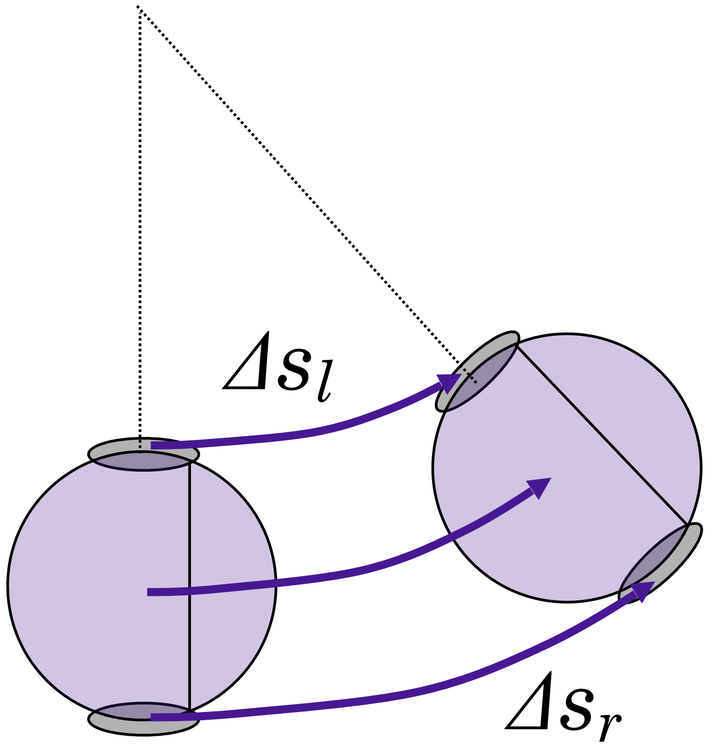
Agenda

1. Kinematics
2. **Differential Drive Robot + Wheel Encoder**
3. Differential Drive Robot + Wheel Velocity

- **Differential drive + Wheel Encoder**
  - If a robot starts from a position p, and the right and left wheels move respective distances Δs_r and Δs_l, what is the resulting new position p' ?



- **Differential drive + Wheel Encoder**
  - To start, let's model the change in angle $\Delta\theta$ and distance travelled Δs by the robot.
    - Assume the robot is traveling on a circular arc of constant radius.

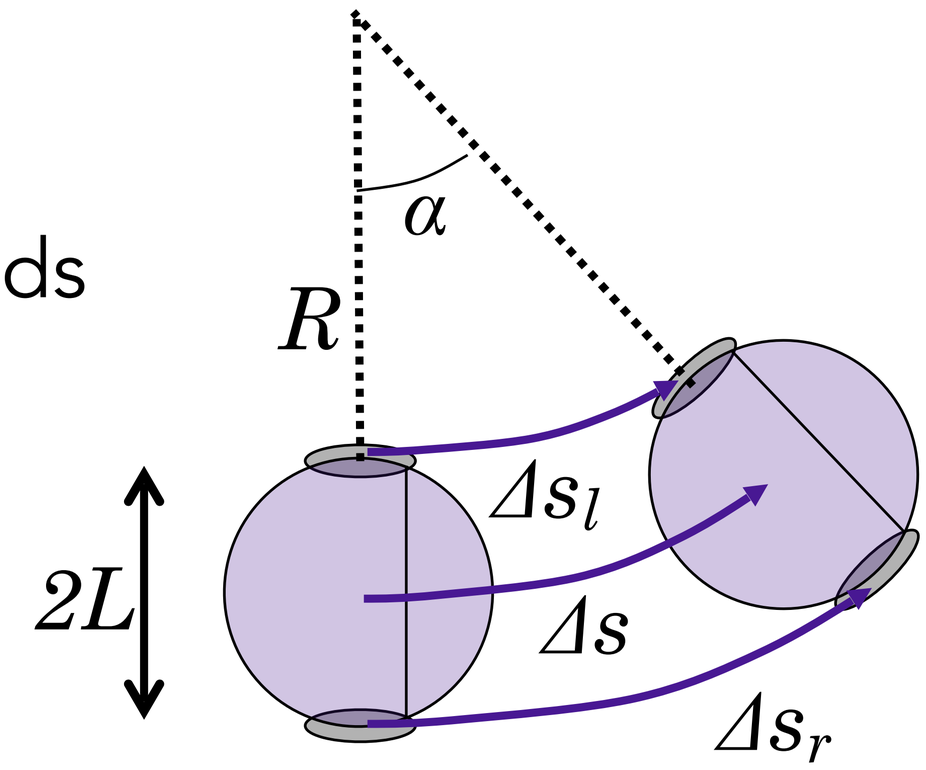


- **Differential drive - Encoder Input**
  - Begin by noting the following holds for circular arcs:

$$\Delta s_l = R\alpha$$

$$\Delta s_r = (R+2L)\alpha$$

$$\Delta s = (R+L)\alpha$$



- **Differential drive - Encoder Input**
  - Now manipulate first two equations:

$$\Delta s_l = R\alpha \qquad \Delta s_r = (R+2L)\alpha$$

- **Differential drive - Encoder Input**
  - Now manipulate first two equations:

$$\Delta s_l = R\alpha \qquad \Delta s_r = (R+2L)\alpha$$

To

$$R\alpha = \Delta s_l$$

$$L\alpha = (\Delta s_r - R\alpha)/2$$

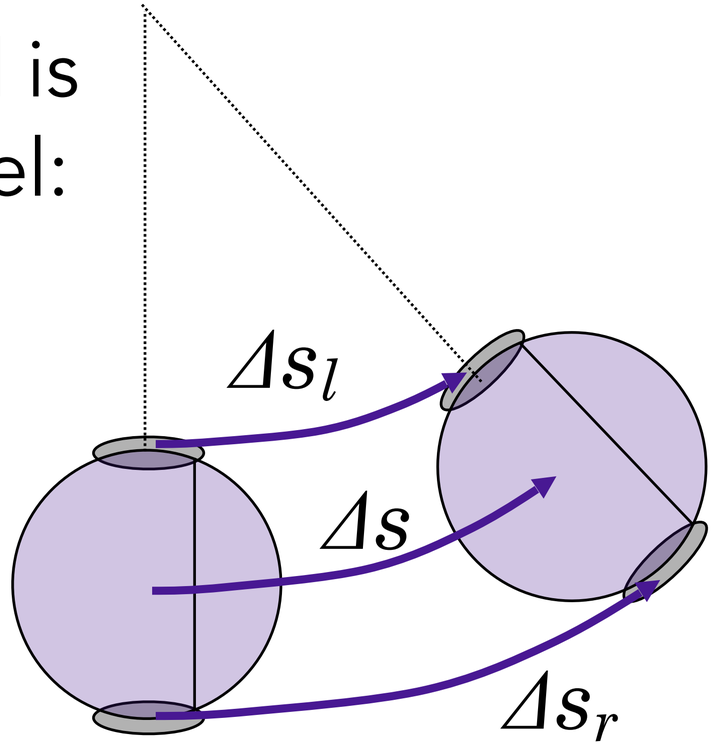
$$= \Delta s_r/2 - \Delta s_l/2$$

- **Differential drive - Encoder Input**
  - Substitute this into last equation for Δs:

$$
\begin{aligned}
\Delta s &= (R+L)\alpha \\
&= R\,\alpha + L\alpha \\
&= \Delta s_l + \Delta s_r\ /2\ -\ \Delta s_l\ /2 \\
&= \Delta s_l\ /2 + \Delta s_r\ /2 \\
&= \frac{\Delta s_l\ + \Delta s_r}{2}
\end{aligned}
$$

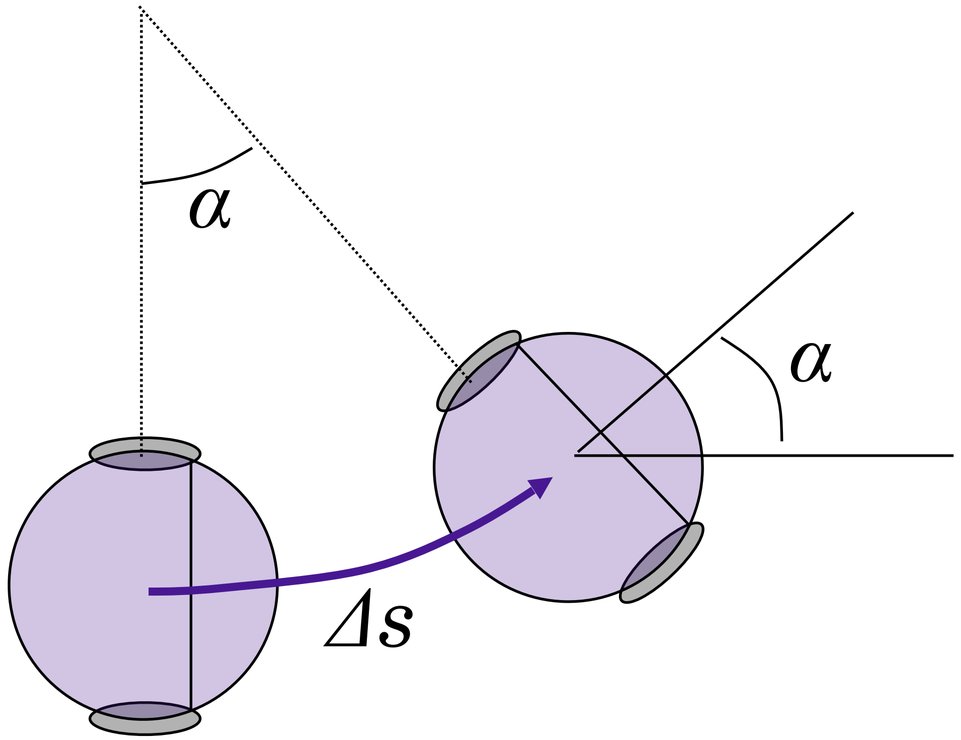
- **Differential drive - Encoder Input**
  - Or, note the distance the center travelled is simply the average distance of each wheel:

$$\Delta s = \frac{\Delta s_r + \Delta s_l}{2}$$



- **Differential drive - Encoder Input**
  - To calculate the change in angle $\Delta\theta$, observe that it equals the rotation about the circular arc's center point

$$\Delta\theta = \alpha$$



- **Differential drive - Encoder Input**
  - So we solve for α by equating α from the first two equations:

$$\Delta s_l = R\alpha \qquad \Delta s_r = (R+2L)\alpha$$

This results in:

$$\Delta s_l \,/\, R = \Delta s_r \,/\, (R+2L)$$

$$(R+2L)\,\Delta s_l = R\,\Delta s_r$$

$$2L\,\Delta s_l = R\,(\Delta s_r - \Delta s_l)$$

$$\frac{2L\,\Delta s_l}{(\Delta s_r - \Delta s_l)} = R$$

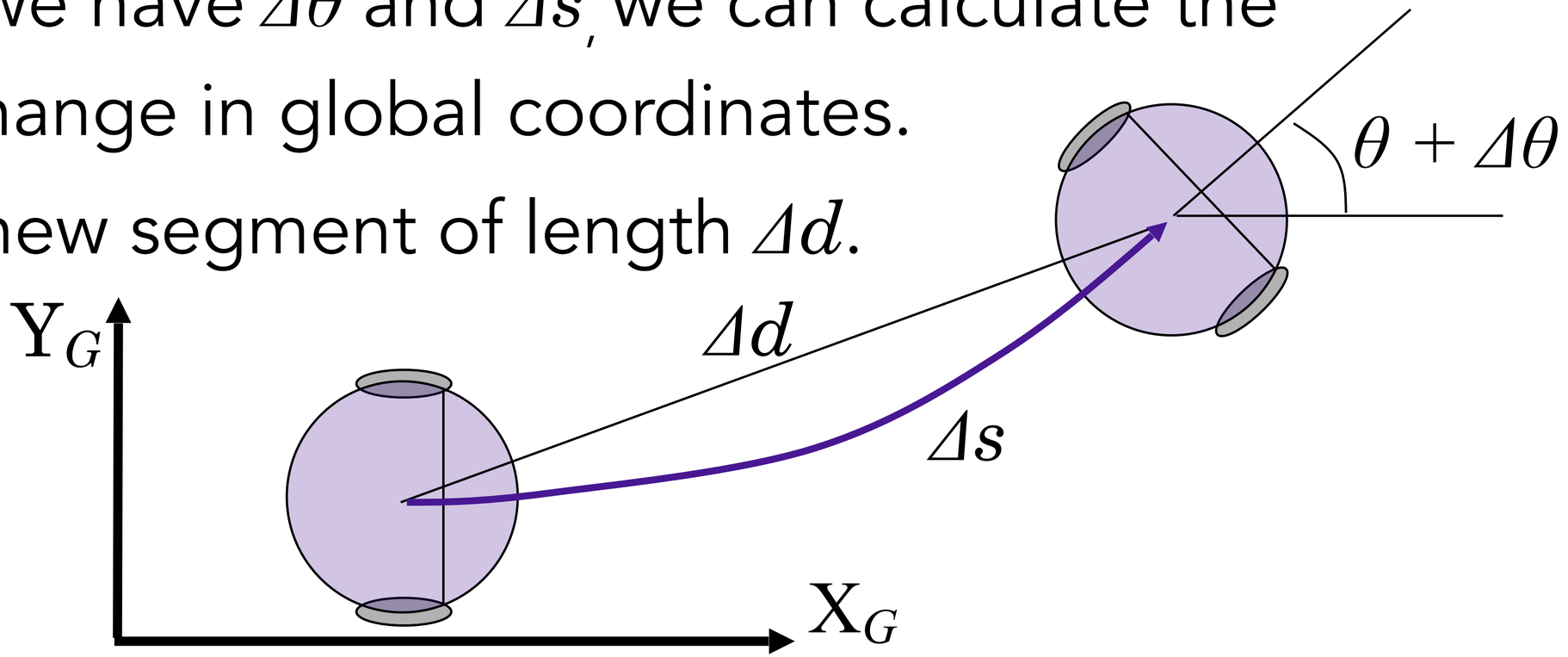
- **Differential drive - Encoder Input**
  - Substitute R into

$$\alpha = \Delta s_l / R$$
$$= \Delta s_l \, (\Delta s_r - \Delta s_l) / (2L \, \Delta s_l)$$
$$= \frac{(\Delta s_r - \Delta s_l)}{2L}$$

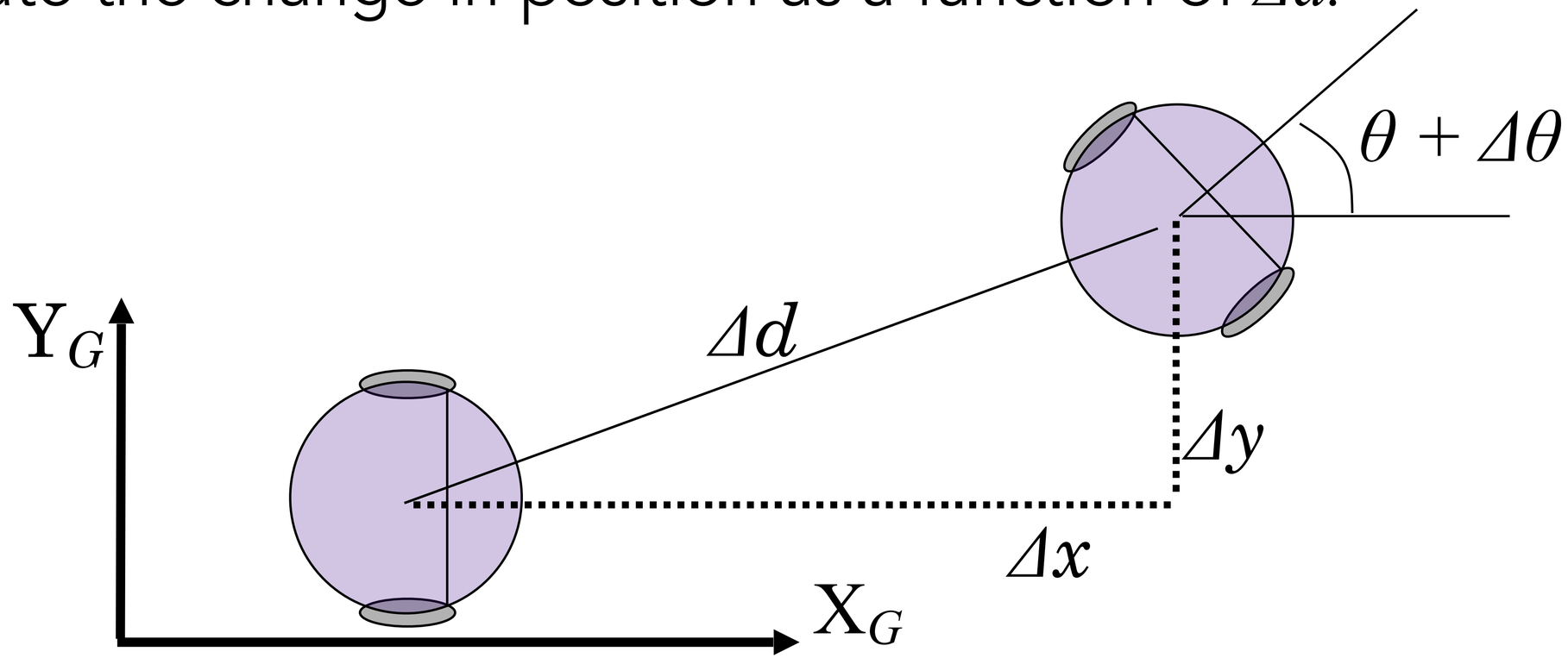
So…

$$\Delta\theta = \frac{(\Delta s_r - \Delta s_l)}{2L}$$

- **Differential drive - Encoder Input**
  - Now that we have $\Delta\theta$ and Δs, we can calculate the position change in global coordinates.
  - We use a new segment of length Δd.



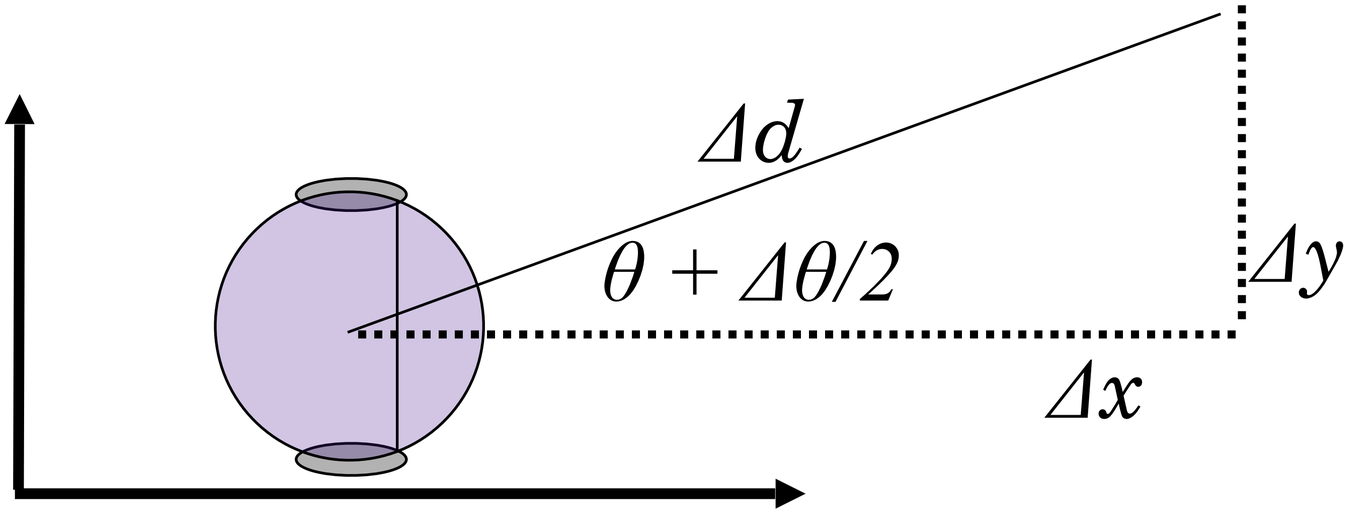
- **Differential drive - Encoder Input**
  - Now calculate the change in position as a function of Δd.



- **Differential drive - Encoder Input**
  - Using Trig:

$$\Delta x = \Delta d \cos(\theta + \Delta\theta/2)$$

$$\Delta y = \Delta d \sin(\theta + \Delta\theta/2)$$



- Differential drive - Encoder Input

$$x_{\mathbf{t}} = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

$$x_t = f(x_{t-1}, u_t)$$

- **Differential drive - Encoder Input**

$$x_{\mathbf{t}} = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

$$x_t = f(x_{t-1}, u_t) = x_{t-1} + \begin{bmatrix} (\Delta s_r + \Delta s_l)cos(\theta_{t-1} + \Delta\theta/2)/2 \\ (\Delta s_r + \Delta s_l)sin(\theta_{t-1} + \Delta\theta/2)/2 \\ (\Delta s_r - \Delta s_l)/(2L) \end{bmatrix}$$

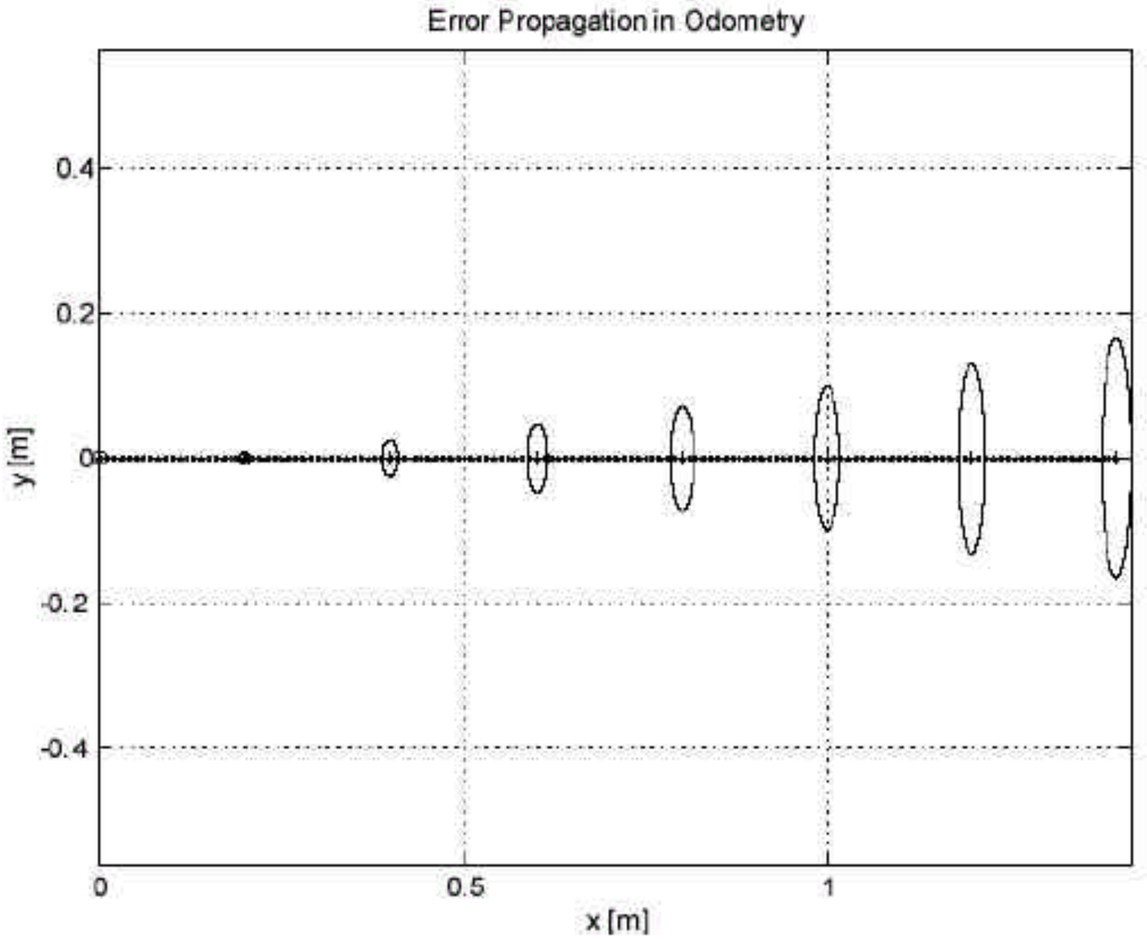
- Differential drive - Encoder Input

$$u_{\mathbf{t}} = \begin{bmatrix} \Delta s_r \\ \Delta s_l \end{bmatrix}$$

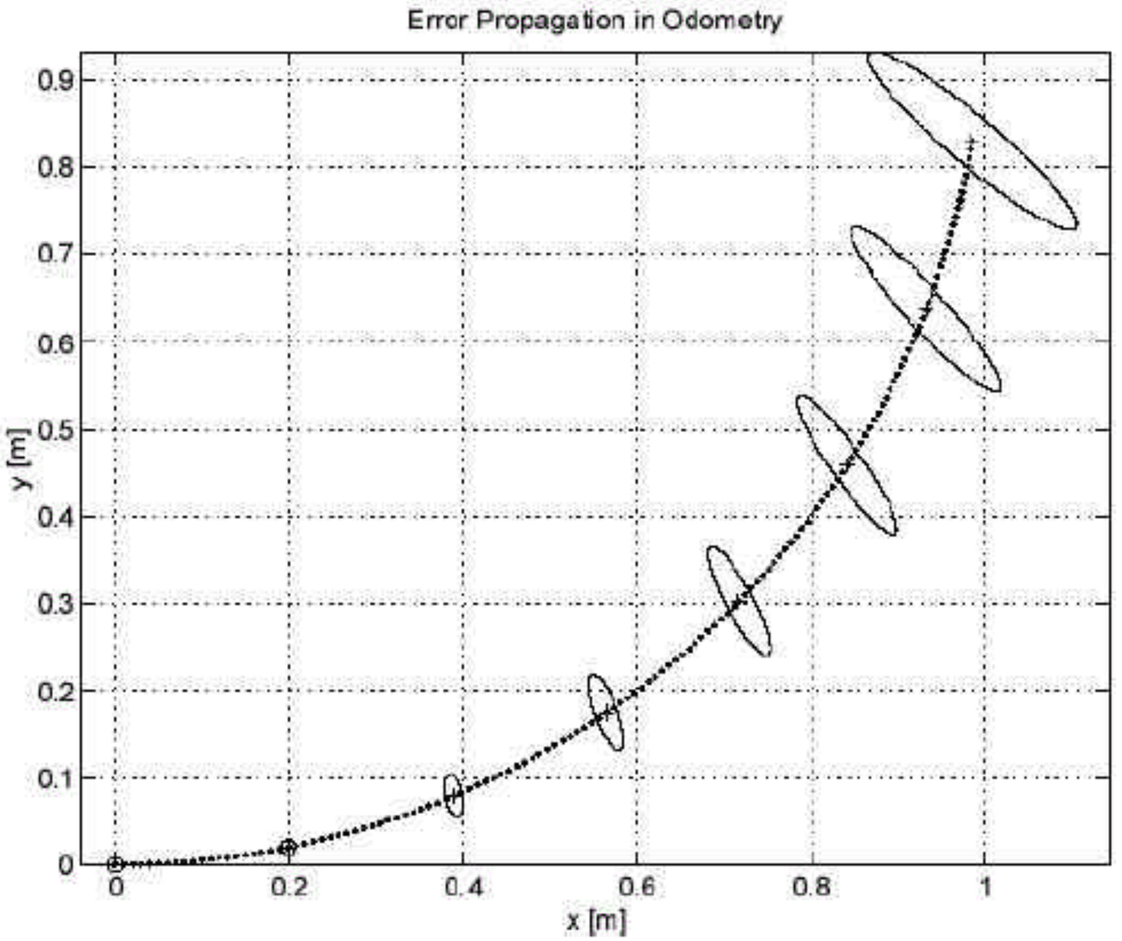
- Differential drive - Encoder Input

$$u_t = \begin{bmatrix} \Delta s_r \\ \Delta s_l \end{bmatrix} + \begin{bmatrix} N(0, \sigma^2_{\Delta s}) \\ N(0, \sigma^2_{\Delta s}) \end{bmatrix}$$

- **Differential drive - Encoder Input**
  - Error propagation



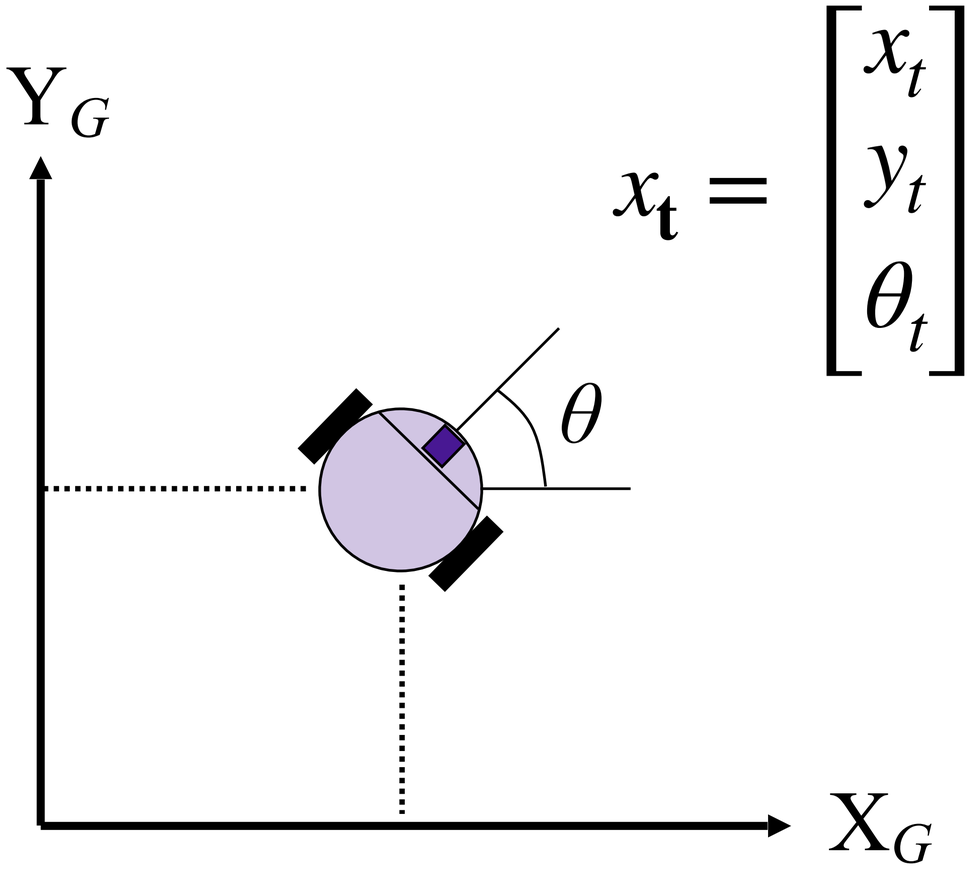
- **Differential drive - Encoder Input**
  - Error propagation



Agenda

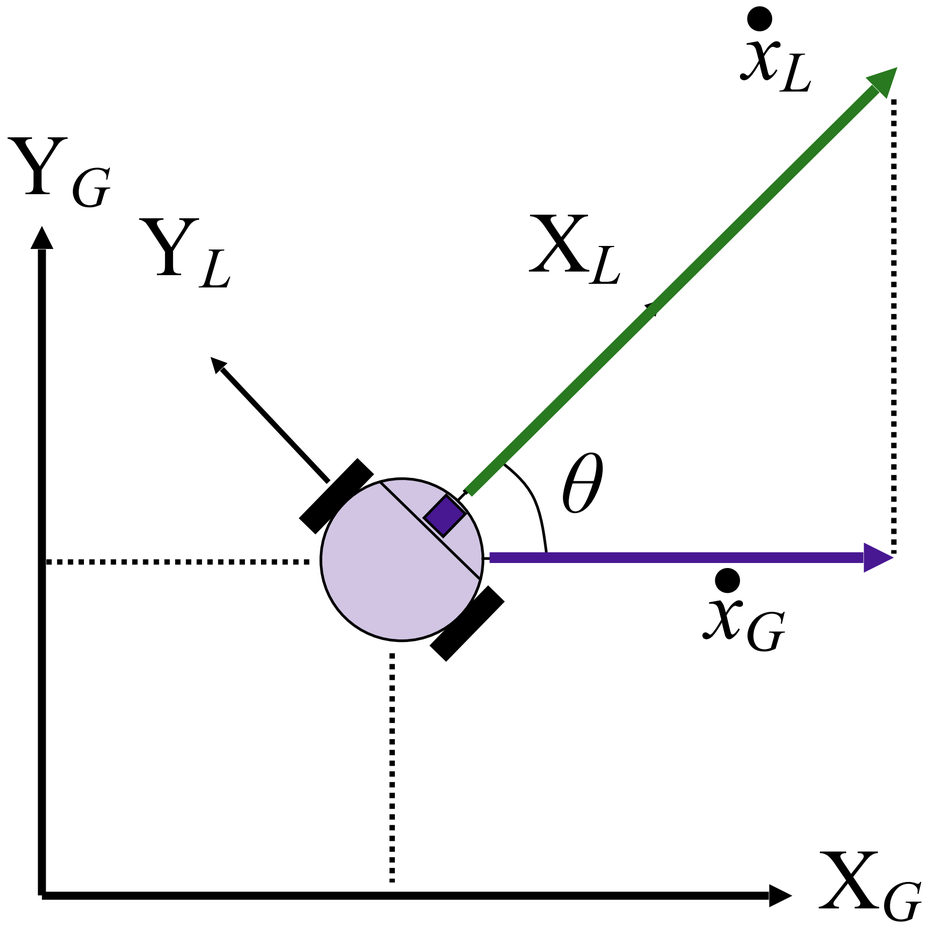
1. Kinematics
2. Differential Drive Robot + Wheel Encoder
3. **Differential Drive Robot + Wheel Velocity**

- **Unicycle Model**
  - Assume perfect control of forward velocity v and rotational velocity w.
  - The motion of a robot can be described as function of v, w.



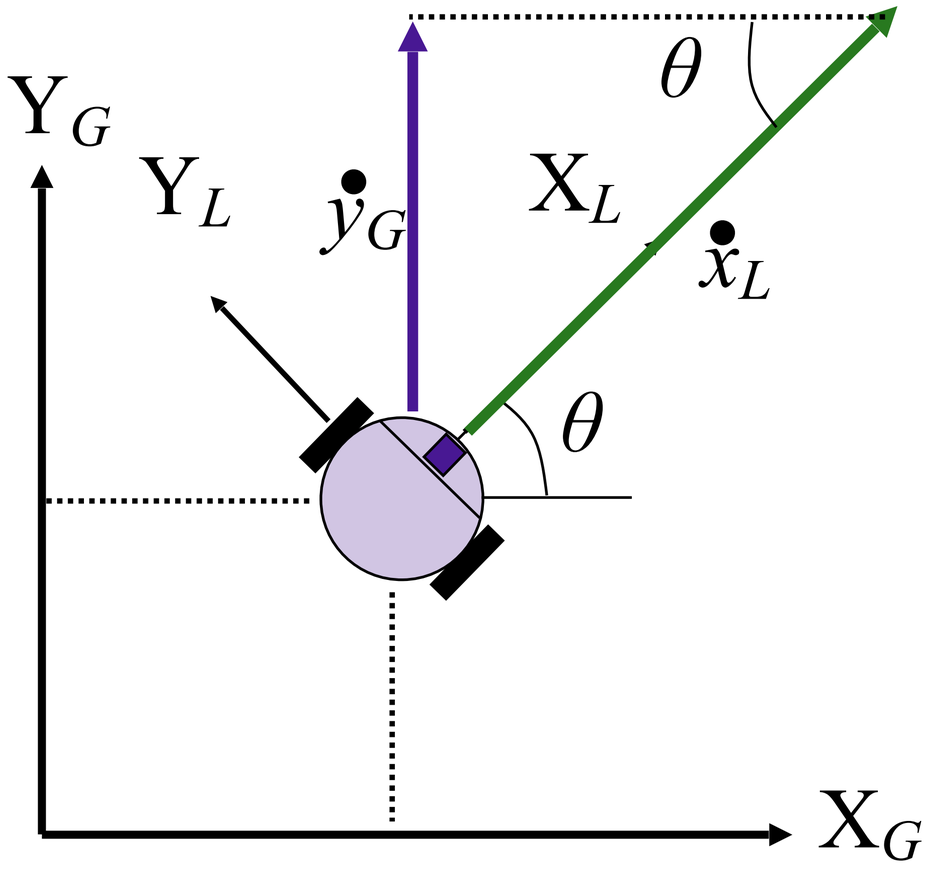
- **Unicycle model**
  - Let's obtain the transformation matrix, starting with the X_G direction:

$$\dot{x}_G = \dot{x}_L cos(\theta)$$



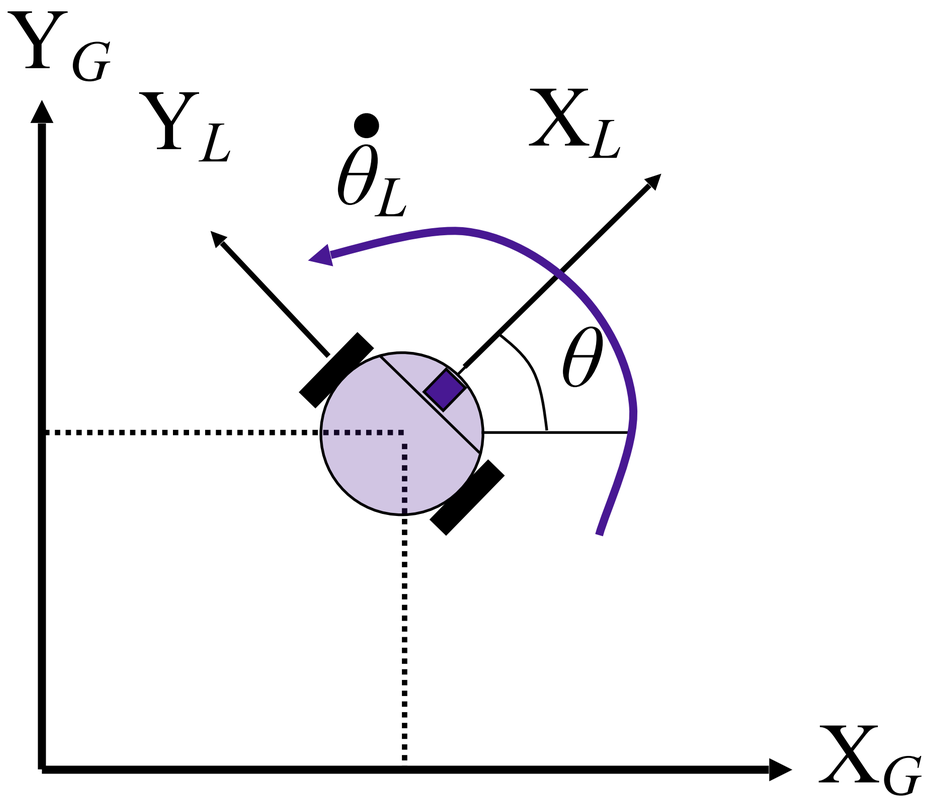
- **Transformations**
  - Let's obtain the transformation matrix, starting with the X_G direction:

$$\dot{y}_G = \dot{x}_L sin(\theta)$$



- **Transformations**
  - What about rotational velocity?

$$\dot{\theta}_G = \dot{\theta}_L$$



- **Transformations**
  - Lets put our equations in matrix form:

$$\begin{bmatrix} \dot{x}_G \\ \dot{y}_G \\ \dot{\theta}_G \end{bmatrix} = \begin{bmatrix} cos(\theta) & 0 & 0 \\ sin(\theta) & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x}_L \\ \dot{y}_L \\ \dot{\theta}_L \end{bmatrix}$$

- **Transformations**
  - Rewriting

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} cos(\theta) & 0 \\ sin(\theta) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

- **Transformations**
  - Rewriting, in discrete time

$$\begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} = \begin{bmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{bmatrix} + \begin{bmatrix} cos(\theta)\Delta t & 0 \\ sin(\theta)\Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

- Differential drive - Encoder Input

$$x_{\mathbf{t}} = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

$$x_t = f(x_{t-1}, u_t)$$

- **Differential drive - Encoder Input**

$$x_{\mathbf{t}} = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

$$x_t = f(x_{t-1}, u_t) = x_{t-1} + \begin{bmatrix} cos(\theta_{t-1})\Delta t & 0 \\ sin(\theta_{t-1})\Delta t & 0 \\ 0 & \Delta t \end{bmatrix} u_t$$

- Differential drive - Encoder Input

$$u_{\mathbf{t}} = \begin{bmatrix} v_t \\ w_t \end{bmatrix}$$

- Differential drive - Encoder Input

$$u_t = \begin{bmatrix} v_t \\ w_t \end{bmatrix} + \begin{bmatrix} N(0,\sigma_v^2) \\ N(0,\sigma_w^2) \end{bmatrix}$$